

ECON712 Replication Exercise

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Topic: Does a Carbon Tax Reduce Co2 Emissions? Evidence from British Columbia

Summary and Contribution

13/15

- **Research question:** The central question of the paper, to quantify the impact of the BC Carbon Tax on total emissions, is clearly stated.
- **Contribution:** Your replication exercise discusses the two contributions of the paper to the literature. A nice addition to the replication exercise would have been to critically analyze these contributions. For instance, I understand that, from a policy perspective, the aggregate effect of the tax on emissions is an important measure to justify the implementation of the policy. However, including sectors with low energy use may add noise to the estimate rather than improve identification of the tax's effect on emissions.
- **Main findings:** Good discussion of the main findings of the paper, including the statement that results should be interpreted with caution as the treatment could correlate with trends due to non-random assignment of the tax across Provinces.

Identification Strategy

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- **Identification Problem:** Your discussion of the identification issue and how it might be resolved using a differences-in-differences identification approach is good. One minor concern with your discussion is that you mention provinces may exhibit differential underlying trends driven by pre-existing dynamics. You need to be careful here as DiD does not solve this issue (unless you use additional controls such as linear trends).
- **Identification Strategy:** Good discussion of DiD, including limitations of this identification approach.
- **Empirical Specification:** Your critical review would have benefited from a more thorough discussion of the empirical specification. For instance, it is interesting that the author included lagged emissions as a dependent variable and used an infinites series approach to combine coefficients and obtain the long-run effect of the policy, as opposed to using long-differences and/or an event study graph. As you correctly point out, this approach requires strict exogeneity of regressors, which in a diff-in-diff framework is hard to justify. Also, the fact that the author introduces contemporaneous GDP, means that they want to capture the effect of the carbon tax on emissions independently of its effect on the aggregate economy. That is, the paper wants to capture the effect that works through substitution towards other energy sources or the decrease in emissions per unit of output.
- **Critical Assessment:** Your critical assessment of the identification strategy is good. A small comment is that you need to be careful in suggesting that introducing prices of other energy product as controls could improve the analysis. The inclusion of these prices could lead to a 'bad control' problem as they likely respond to the introduction of the carbon tax.

Replication

37/40

- **Data Description:** Good discussion of the data and sources. Is the data in Table 1 annual? —I could not find that information in your paper.
- **Main Results:** Good presentation of main results. You did a very good job replicating the paper. Small issue, you cannot write your S.E. in percentages (it is fine to multiply the SE by 100 when you write your coefficients in percentages, but you cannot write the SE as percentages – unless you are calculating the coefficient of variation). Good discussion of additional robustness in the Appendix. On the other hand, Figure III should have been either excluded from your review or discussed more thoroughly. It is not clear from your replication exercise how the analysis is performed.

Critical Discussion and Proposed Extensions

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- **Assessment:** Correct assessment of empirical choices and limitations of the analysis.
- **Proposed Improvements:** Some of your proposed improvements are valid, including a firm-level analysis and the synthetic control approach, which the paper ultimately relies on to show the effect of the carbon tax. An issue with the heterogeneity analysis you are suggesting is that it is infeasible with a single treated unit.

Grade: 90/100

ECON712 Presentation

- **Quality, organization, and clarity of your presentation (1.5 /1.5) :**The presentation was well organized with a very good flow. Starting with the outline of your presentation was great as it provided a roadmap for your entire presentation.
- **Summary of the paper and general discussion of the paper's contribution to the literature (1.5 /2)**
 - Good overview of the paper, including the two contributions of the paper and a preliminary discussion of results and limitations of the empirical analysis
 - The discussion of the Did could have been clearer. For instance, instead of mentioning that the identification strategy includes fixed effects, a holistic discussion of the empirical benefits of using fixed effects would have improved your presentation.
- **Discussion of the data, methods, results, and identification-related issues (4 /4.5)**
 - Good introduction of data and data sources used in the replication exercise.
 - The main estimating equation could have been presented better. For instance, you did not discuss in detail why each control is included in the regression equation.
 - Properly presented the dynamic trends graphs and discussed implications for main results.
 - Provided an adequate discussion of the paper's results.
- **Overall assessment of the paper including suggested improvements (1.5 /1.5)**
 - Good assessment of the strengths and limitations of the paper.
- **Ability to respond to questions and your overall conclusion, including a brief discussion of whether the paper increased your interest in the topic (0.5 /0.5)**

Grade: 9/10